Measure of Dispersion

Important Questions:-

Very short type:-

(1) (a) If A.M and C.V of a data on a variable x are 10 and 50%. Find the standard deviation of $(3-2x)$ [Ans:- 10]

(b) Find the GM of the set of observation for which A.M = 2. and S.D = 0 [Ans - 2]

(c) Write down a measure of relative dispersion [Ans:- Coefficient of Range]

(d) Find out the standard deviation of two real number a and b. [Ans: $\frac{|a-b|}{\sqrt{2}}$]

(e) Prove or disprove: $\sum_{i=1}^{50} |i-25.1| = \sum_{i=1}^{50} |i-25.2|$

[Ans :- 625 (equal)]

(f) The S.D of 15 item is 6. If each item is increased by 2. then new S.D will be ? [Ans 6]

(g) The S.D of identical units ie 2,2,2,2,2 is ? [Ans :- 0]

(h) If $n = 100$, $\bar{x} = 2$, $\sigma = 4$, Find the value of $\sum x$ and $\sum x^2$? [Ans:- 200, 2000]

(i) If $n = 20$, $\sum x = 6$ and $\sum x^2 = 821$, then Find the value of mean and S.D. [Ans:- $\sqrt{\frac{4094}{100}}$]

(j) Two samples of size 40 and 50 respectively 40 and 50 have same mean but different S.D 19 and 8 respectively. Find the S.D of Combined sample of 90.

[Ans:- $\approx$ 13]

(k) The A.M. of the ~~sum~~ run scored by the three batsmen, Ram, Shyam, and Sohan in the same series of 10 innings are 50, 48 and 12 respectively. The S.D of their runs are 15, 12 and 2 respectively. Find the most consistent batsman. [Ans:- Sohan most consistent]

(l) The mean of 100 observations is 25. and their S.D σ is 9. If 6 is substract from each of the observations, then Find the mean and S.D. (19, 9 - Ans)

(m) The sum of squares of deviation of the number 8, 3, 5, 12 and 10 from any number 'a' is minimum. Then find the value of 'a'.

[Ans:- 7.6]

Short/ Long Answer type:—

(1) What is Coefficient of Variation. Write its uses.

(2) Discuss merit and demerit of standard deviation as a measure of dispersion

(3) What do you mean by dispersion of a data set? Suggest suitable measure of dispersion to compare heights of two different ethnic groups.

(4) Obtain standard deviation of first n natural numbers.

Theorems:—

(1) Let there be two groups n_1 and n_2 with values with mean $\bar{x}_1$, $\bar{x}_2$ and variances s_1^2 and s_2^2 respectively. Then show that Combine variance s^2 of (n_1+n_2) values can be expressed as

$$s^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1+n_2} + \frac{n_1 n_2}{(n_1+n_2)^2}(\bar{x}_1-\bar{x}_2)^2$$

(2) For a set of n observations show that $s^2 \leq \frac{R^2}{4}$

where s = standard deviation and
R = Range of given observations

Questions:-

5. A group of Students took two exams. The mean score for Exam 1 was 70 with a standard deviation of 5 and the mean score for Exam 2 was 75 with standard deviation of 8. Which exam had greater variability and why?

6. Explain how box plots can be used to visualize measure of dispersion and skewness